THERMITES

We are thermites, we are swarm

INTRODUCTION

Problem analysis

What do we need to achieve?

Breakdown

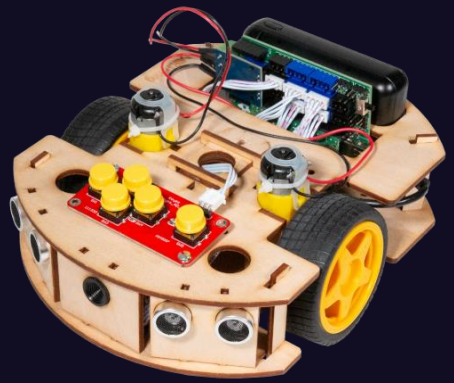
How can we achieve them?

Define

What do we need to implement?

PROBLEM

- Deliver items to objectives
  - In a confined, unknown environment
- One mobile manipulator
- Four mobile robots with sensors
  - Camera
  - Ultrasonic sensor



CRITERIA

#	Criterion	Key threshold
1	Stay in bounds	0 Boundary violation
2	AprilTag detection range	≥ 1.0 m, ±10 cm accuracy
3	Object stays in basket	100% inside basket boundary
4	No collisions	0 contact events
5	No inter-team interference	0 obstruction events
6	Full stop on signal	Stops within 3 seconds
7	Autonomous operation	No human intervention

SCENARIO BREAKDOWN

- Exploration
- Objective location estimation

- Item pick and place
- Objective navigation

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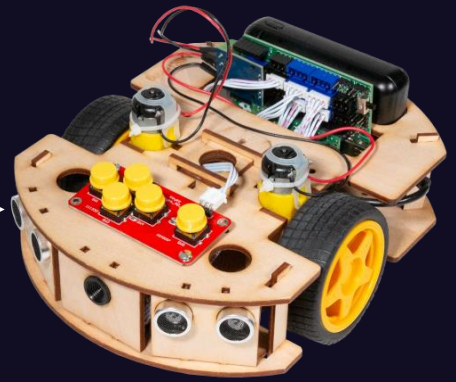
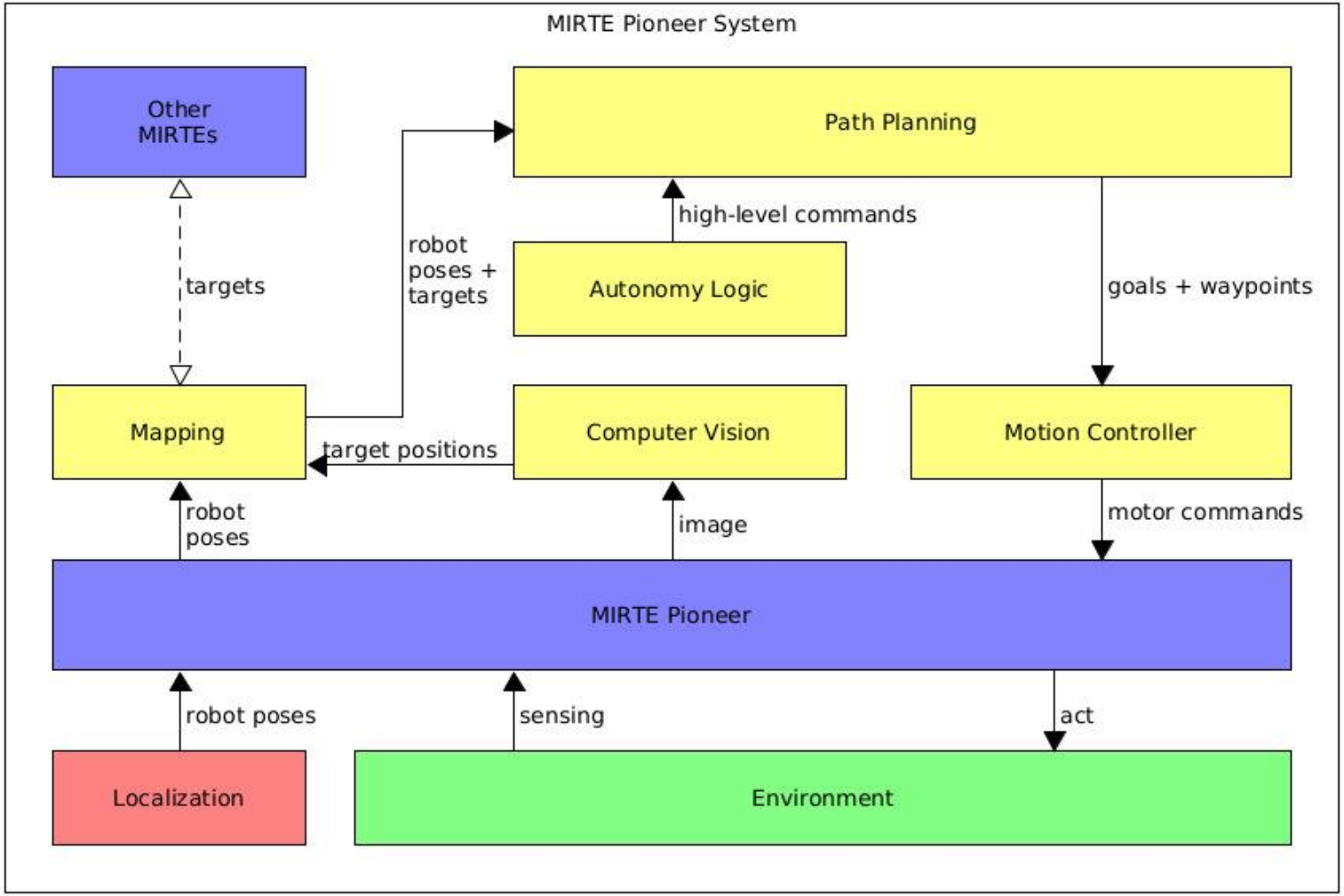
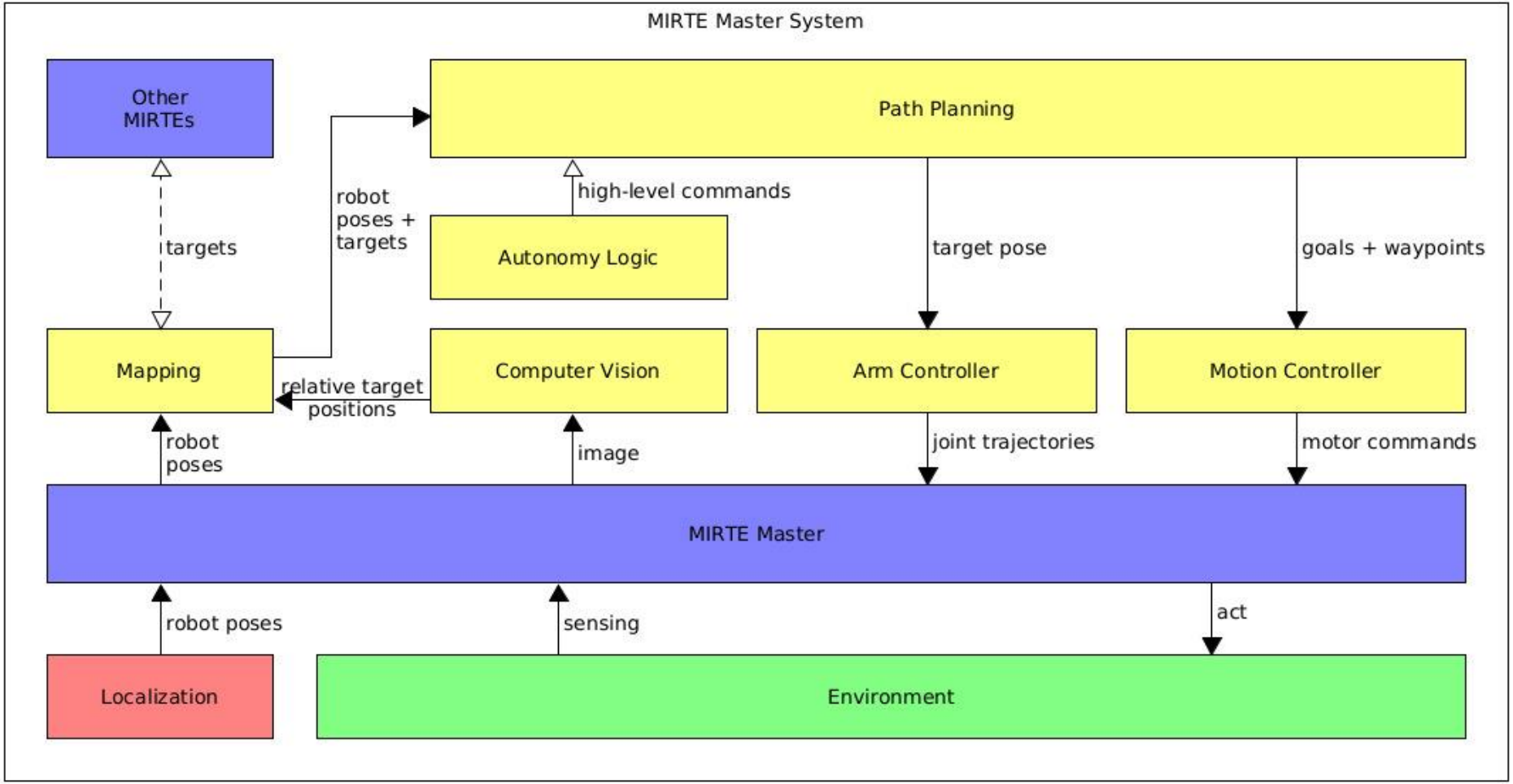
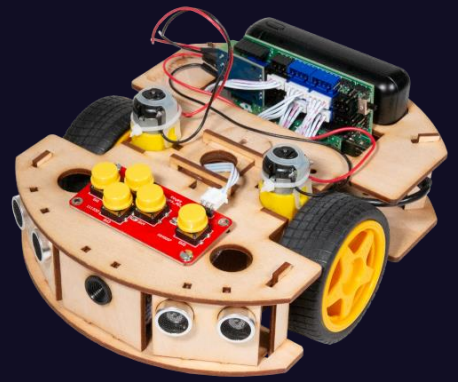


- Item pick and place
- Objective navigation

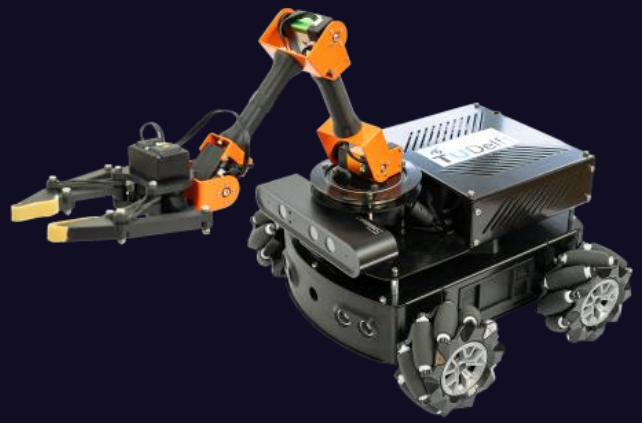

MIRTE Pioneer System
Other MIRTEs
Path Planning
high-level commands
robot poses + targets
Autonomy Logic
goals + waypoints
targets
Mapping
Computer Vision
Motion Controller
target positions
robot poses
image
motor commands
MIRTE Pioneer
robot poses
sensing
act
Localization
Environment


MIRTE Master System
Other MIRTEs
Path Planning
targets
robot poses + targets
high-level commands
Autonomy Logic
target pose
goals + waypoints
Mapping
relative target positions
Computer Vision
Arm Controller
Motion Controller
robot poses
image
joint trajectories
motor commands
MIRTE Master
robot poses
sensing
act
Localization
Environment

SWARMING STRATEGY

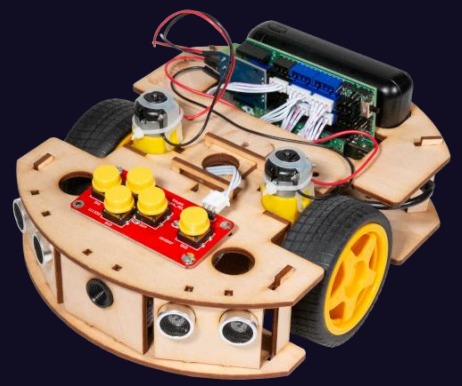


- Explore
- Publish objectives' locations

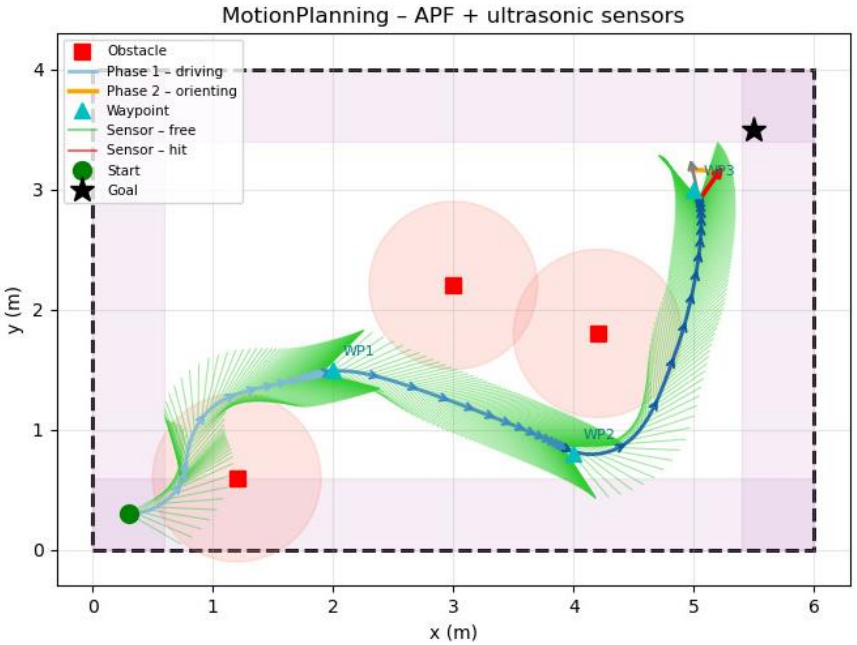


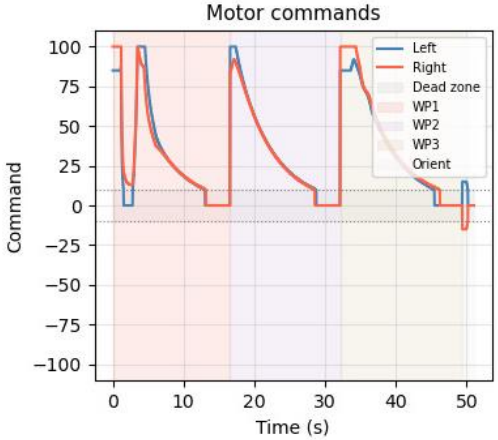
- Focus on objectives
- Navigating to and from objectives
- Pick and place items

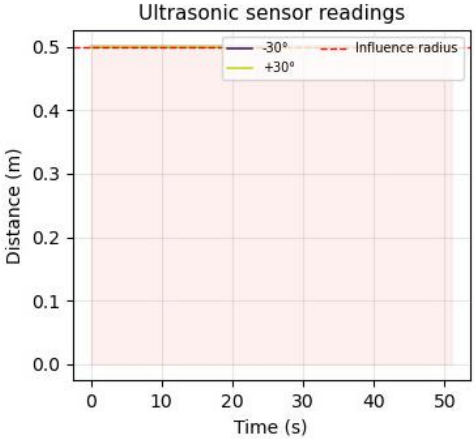
EXPLORATION



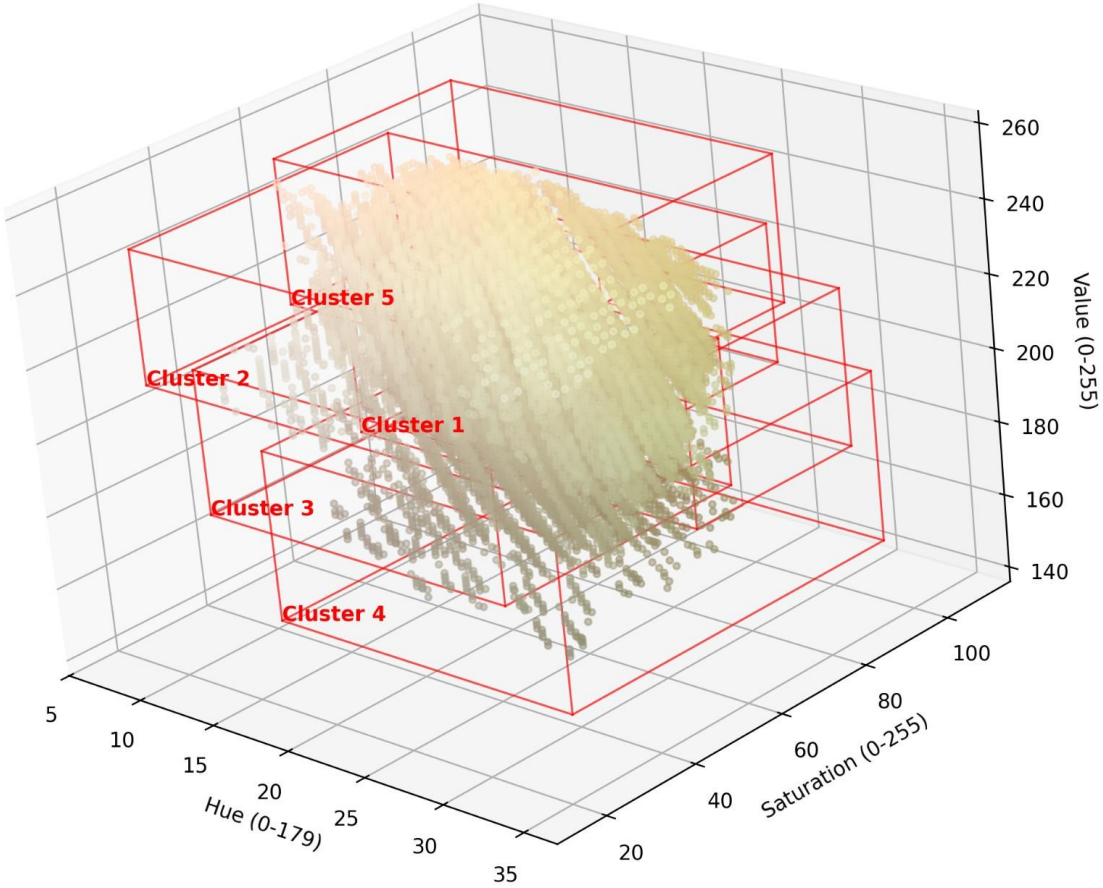
- Divide the map into 4 quadrants
- Assign one robot to each quadrant
- Motion planning with obstacle avoidance
  - Create artificial potential field
  - Using ultrasonic sensors
- Cost function for path planning
  - Promotes navigation towards visible baskets
  - Minimizes the distance to them
  - Penalizes switching targets
- Broadcast tag and current position
- Estimate object position and publish to master
- Publish also obstacle position to master
  - Filter out robots

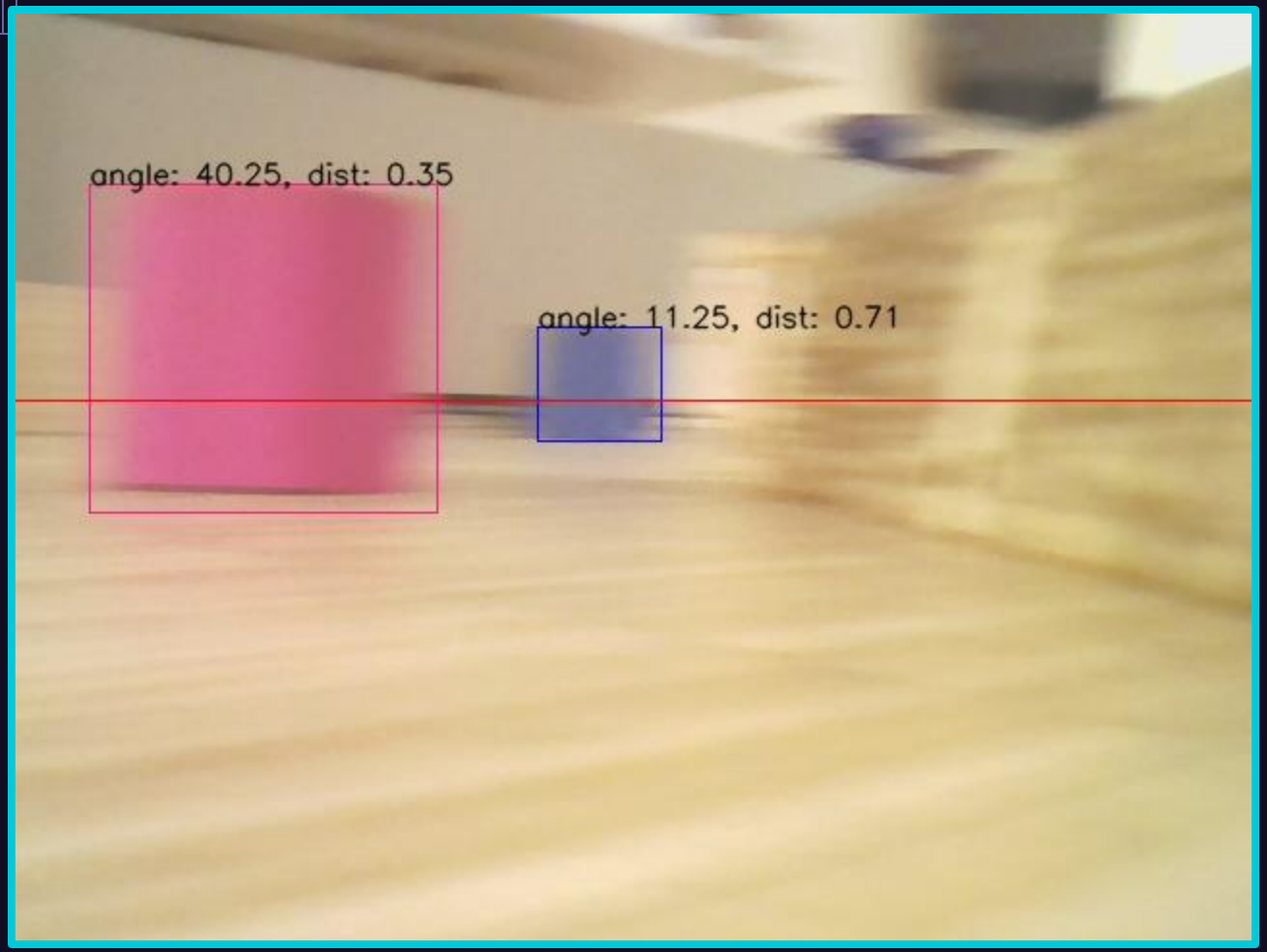

MotionPlanning – APF + ultrasonic sensors
Obstacle
Phase 1 – driving
Phase 2 – orienting
Waypoint
Sensor – free
Sensor – hit
Start
Goal
WP1
WP2
WP3
y (m)
x (m)


Motor commands
Left
Right
Dead zone
WP1
WP2
WP3
Orient
Command
Time (s)


Ultrasonic sensor readings
-30°
+30°
Influence radius
Distance (m)
Time (s)

Top HSV Values and their Clusters

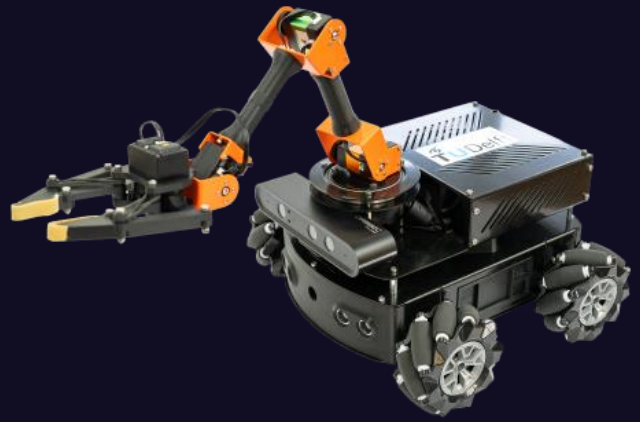



angle: 40.25, dist: 0.35
angle: 11.25, dist: 0.71

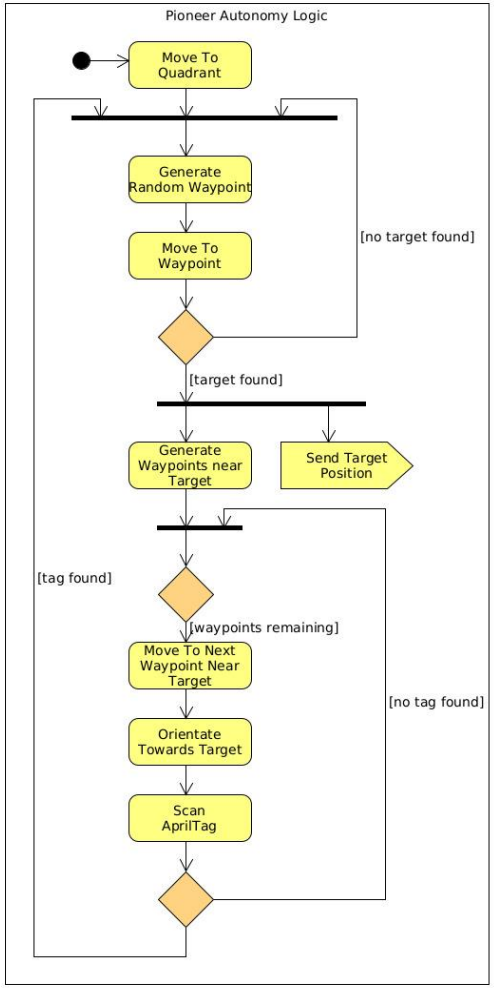
NAVIGATION

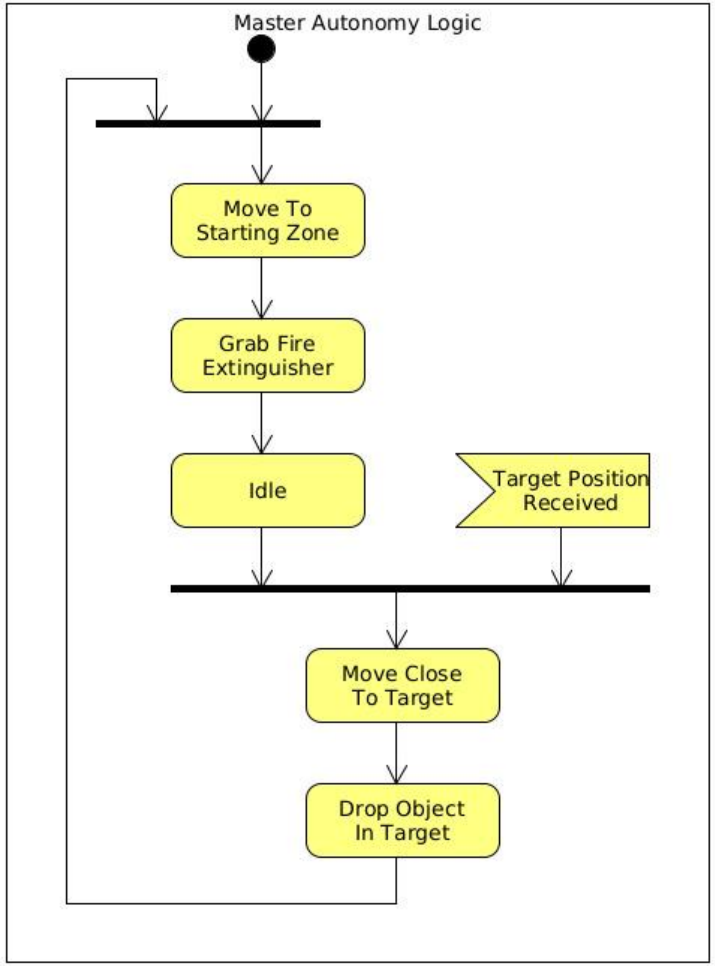
- Create map using data from Pioneers
  - 2D occupancy grid map
- Receive objectives from Pioneers
  - Stored in buffer
  - Sorted by distance to objective
  - Pop the first one whenever idle
- Run A* to find optimal path to selected the objective
- Keep track of mission progress
  - Filter out objectives already done

GRASPING AND DROPPING

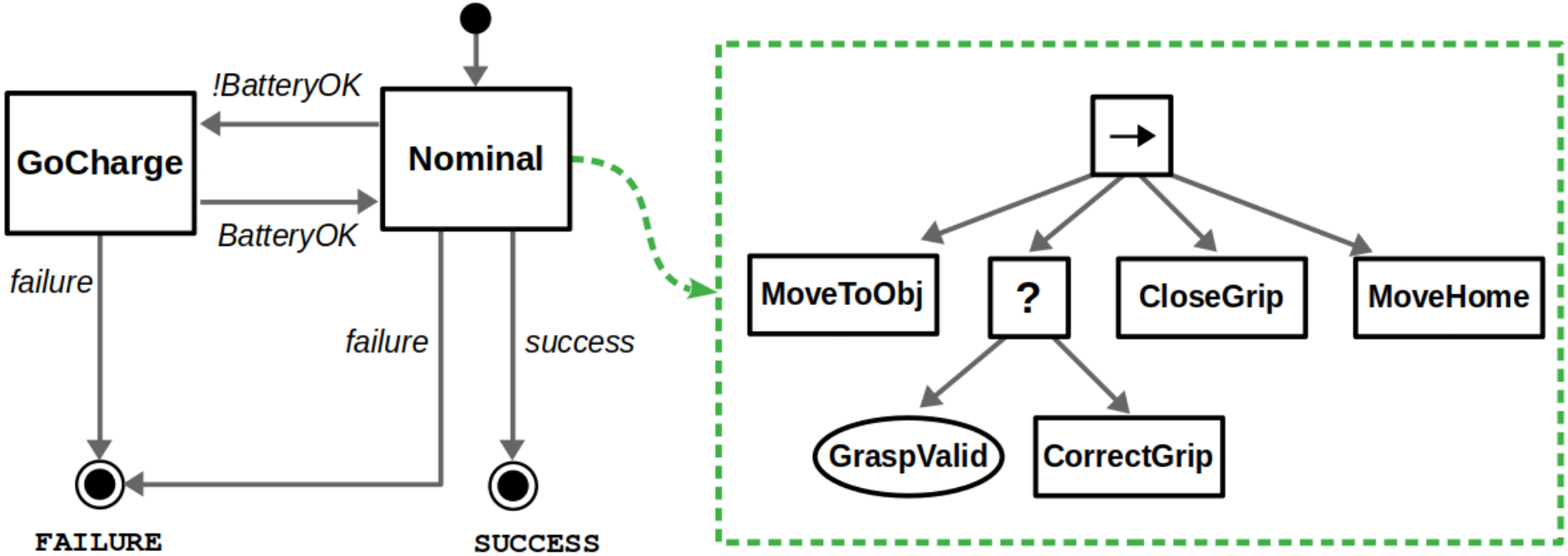


- Use RGBD camera to align robot position
- Estimate item position
- Transform to gripper frame
- Perform Inverse Kinematics to compute grasp/place pose
- If IK fails, adjust robot position


Pioneer Autonomy Logic
Move To Quadrant
Generate Random Waypoint
Move To Waypoint
[no target found]
[target found]
Generate Waypoints near Target
Send Target Position
[tag found]
[waypoints remaining]
Move To Next Waypoint Near Target
[no tag found]
Orientate Towards Target
Scan AprilTag


Master Autonomy Logic
Move To Starting Zone
Grab Fire Extinguisher
Idle
Target Position Received
Move Close To Target
Drop Object In Target

MISSION DIRECTING



- Hierarchical State Machine and Behaviour Tree hybrid
- Modular and robust behaviours
- Safety guarantee